Chapter 4. Systematic Sampling

Contents

1 Introduction **1**

2 Selection Methods **2**

2.1 Method A: random start among the first k . . . 2

2.2 Method B: random start among all N . . . 2

3 Estimation of the Population Mean **2**

3.1 Unbiasedness . . . 2

3.2 Variance of $\bar{y}_{\text{sy}}$ $(N = nk)$ . . . 3

3.3 Intraclass correlation . . . 3

3.4 ANOVA form . . . 4

3.5 Types of population . . . 5

3.6 Estimating the variance . . . 5

4 Population Total and Proportion **5**

5 Selecting the Sample Size **5**

6 Selected exercises **6**

1 Introduction

Picking names at equal intervals from a random start gives a systematic sample.

Definition 4.1

A **1-in-k systematic sample with a random start** takes a random one of the first k frame elements and every kth thereafter; n is the sample size, k the number of possible samples.

Advantages over SRS:

- the sample is easy to draw and the field work is less prone to mistakes;
- it can provide more information per unit cost, because the sample is spread evenly over the frame.

2 Selection Methods

2.1 Method A: random start among the first k

Example 4.2: $N = 12$, $k = 3$

Picking y_2 among y_1, y_2, y_3 gives y_2, y_5, y_8, y_{11}: the population is cut into $n = N/k = 4$ rows, one unit per row, and the $k = 3$ samples are the columns

#1	#2	#3
y_1	y_2	y_3
y_4	y_5	y_6
y_7	y_8	y_9
y_{10}	y_{11}	y_{12}

In general the sample is $j,\ j+k,\ j+2k,\ \ldots,\ j+(n-1)k$ with j uniform on $\{1,\ldots,k\}$, and each systematic sample has probability $1/k$.

If $N \neq nk$ the sizes differ: $N = 12$, $k = 5$ give (y_1, y_6, y_{11}), (y_2, y_7, y_{12}), (y_3, y_8), (y_4, y_9), (y_5, y_{10}), each with probability $1/5$.

2.2 Method B: random start among all N

Select j at random from the whole population; with remainder r of j by k, start at y_r ($r = 0$: y_k) and take every kth unit.

Example 4.3: $N = 11$, $k = 3$

If $j = 8$, then $8 = 2 \cdot 3 + 2$, so $r = 2$, the start is y_2 and the sample is (y_2, y_5, y_8, y_{11}). The possible samples are

$$\#1 = (y_1, y_4, y_7, y_{10}), \quad \#2 = (y_2, y_5, y_8, y_{11}), \quad \#3 = (y_3, y_6, y_9).$$

Sample #2 is obtained whenever $j \in \{2, 5, 8, 11\}$, so its probability is $4/11$; similarly #1 has probability $4/11$ and #3 has $3/11$.

Remark 4.4: A versus B

Same samples, different probabilities: $1/k$ (Method A) vs n_i/N (Method B, needing N); equal when $N = nk$.

3 Estimation of the Population Mean

Let y_{ij} be the jth element of the ith systematic sample ($i = 1,\ldots,k$, $j = 1,\ldots,n_i$), $\bar{y}_i = \frac{1}{n_i}\sum_j y_{ij}$, and $\bar{y}_{\text{sy}}$ the mean of the sample actually selected.

3.1 Unbiasedness

Case 1 (Method A, $N = nk$).

$$\mathrm{E}(\bar{y}_{\text{sy}}) = \frac{1}{k}(\bar{y}_1 + \cdots + \bar{y}_k) = \frac{1}{k}\frac{1}{n}(y_1 + \cdots + y_N) = \mu,$$

so $\bar{y}_{\text{sy}}$ is unbiased.

Example 4.5

Population $6,3,4,9,2,5,1,7,8$ with $k=3$: the samples are $(6,9,1)$, $(3,2,7)$, $(4,5,8)$ with means $16/3$, $12/3$, $17/3$, whose average is $5=\mu$.

Case 2 (Method A, $N \neq nk$). For $y_1,\dots,y_7$ and $k=3$,

$$\mathrm{E}(\bar{y}_{\text{sy}}) = \frac{1}{3}\Big[\frac{y_1+y_4+y_7}{3} + \frac{y_2+y_5}{2} + \frac{y_3+y_6}{2}\Big] \neq \frac{y_1+\cdots+y_7}{7} = \mu$$

in general: $\bar{y}_{\text{sy}}$ is *biased*.

Case 3 (Method B, $N \neq nk$). With probabilities $3/7, 2/7, 2/7$,

$$\mathrm{E}(\bar{y}_{\text{sy}}) = \frac{3}{7}\cdot\frac{y_1+y_4+y_7}{3} + \frac{2}{7}\cdot\frac{y_2+y_5}{2} + \frac{2}{7}\cdot\frac{y_3+y_6}{2} = \frac{y_1+\cdots+y_7}{7} = \mu.$$

In general $\mathrm{E}(\bar{y}_{\text{sy}}) = \sum_i \frac{n_i}{N}\bar{y}_i = \frac{1}{N}\sum_i\sum_j y_{ij} = \mu$: Method B is always unbiased.

3.2 Variance of $\bar{y}_{\text{sy}}$ ($N = nk$)

By definition,

$$\mathrm{Var}(\bar{y}_{\text{sy}}) = \frac{1}{k}\sum_{i=1}^{k}(\bar{y}_i-\mu)^2,$$

where $S^2 = \frac{1}{N-1}\sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij}-\mu)^2$.

Theorem 4.6: Variance of the systematic mean

$$\mathrm{Var}(\bar{y}_{\text{sy}}) = \frac{N-1}{N}S^2 - \frac{1}{N}\sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij}-\bar{y}_i)^2.$$

Proof. Decompose the total sum of squares:

$$\sum_i\sum_j(y_{ij}-\mu)^2 = \sum_i\sum_j(y_{ij}-\bar{y}_i)^2 + n\sum_i(\bar{y}_i-\mu)^2 + 2\sum_i(\bar{y}_i-\mu)\sum_j(y_{ij}-\bar{y}_i),$$

and the cross term vanishes because $\sum_j(y_{ij}-\bar{y}_i)=0$. The left side is $(N-1)S^2$ and $n\sum_i(\bar{y}_i-\mu)^2 = nk\mathrm{Var}(\bar{y}_{\text{sy}}) = N\mathrm{Var}(\bar{y}_{\text{sy}})$. Divide by N. ∎

The first term is the population variance, the second the pooled variation *within* systematic samples: heterogeneous samples make $\mathrm{Var}(\bar{y}_{\text{sy}})$ small.

Remark 4.7: Periodicity

For 1 2 3 4 5 | 1 2 3 4 5 | 1 2 3 4 5, $k=5$, every sample is constant, e.g. $(2,2,2)$: no within-sample variation, maximal $\mathrm{Var}(\bar{y}_{\text{sy}})$.

3.3 Intraclass correlation

Homogeneity within a systematic sample is measured by the **intraclass correlation coefficient**

$$\rho = \frac{\mathrm{E}[(y_{ij}-\mu)(y_{ij'}-\mu)]}{\mathrm{E}[(y_{ij}-\mu)^2]}, \qquad j \neq j',$$

where the expectations are averages over the $kn(n-1)/2$ pairs in the same sample and over the N units, respectively.

Theorem 4.8

$$\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = \frac{S^2}{n}\cdot\frac{N-1}{N}[1+(n-1)\rho], \qquad \rho = \frac{2}{(n-1)(N-1)S^2}\sum_{i=1}^{k}\sum_{j<j'}(y_{ij}-\mu)(y_{ij'}-\mu).$$

Derivation

$$\begin{aligned}\mathrm{Var}(\bar{y}_{\mathrm{sy}}) &= \frac{1}{k}\sum_i\Big(\frac{1}{n}\sum_j y_{ij}-\mu\Big)^2 = \frac{1}{n^2k}\sum_i\Big(\sum_j(y_{ij}-\mu)\Big)^2\\ &= \frac{1}{n^2k}\Big[\sum_i\sum_j(y_{ij}-\mu)^2 + 2\sum_i\sum_{j<j'}(y_{ij}-\mu)(y_{ij'}-\mu)\Big].\end{aligned}$$

The denominator of ρ is $\frac{1}{N}\sum_i\sum_j(y_{ij}-\mu)^2 = \frac{N-1}{N}S^2$, and the numerator is $\frac{2}{kn(n-1)}\sum_i\sum_{j<j'}(\cdot)(\cdot)$, so

$$2\sum_i\sum_{j<j'}(y_{ij}-\mu)(y_{ij'}-\mu) = kn(n-1)\frac{N-1}{N}S^2\rho = (n-1)(N-1)S^2\rho,$$

which is the stated formula for ρ. Substituting, with $n^2k = nN$,

$$\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = \frac{1}{nN}[(N-1)S^2 + (n-1)(N-1)S^2\rho] = \frac{S^2}{n}\frac{N-1}{N}[1+(n-1)\rho].$$

Compare with SRS, $\mathrm{Var}(\bar{y}_{\mathrm{ran}}) = \frac{S^2}{n}(1-\frac{n}{N})$:

- ρ large and positive (homogeneous systematic samples) $\Rightarrow \mathrm{Var}(\bar{y}_{\mathrm{sy}})$ large;
- ρ small positive or negative (heterogeneous samples) $\Rightarrow \mathrm{Var}(\bar{y}_{\mathrm{sy}})$ small;
- $\rho \approx 0 \Rightarrow \mathrm{Var}(\bar{y}_{\mathrm{sy}}) \approx \mathrm{Var}(\bar{y}_{\mathrm{ran}})$. Indeed $\rho = -\frac{1}{N-1}$ reproduces the SRS variance exactly.

3.4 ANOVA form

Arrange the population as an $n \times k$ table whose columns are the systematic samples, with column means $\bar{y}_1, \ldots, \bar{y}_k$ and grand mean μ. Let

$$SS_T = \sum_i\sum_j(y_{ij}-\mu)^2, \quad SS_W = \sum_i\sum_j(y_{ij}-\bar{y}_i)^2, \quad SS_B = n\sum_i(\bar{y}_i-\mu)^2,$$

so $SS_T = SS_W + SS_B$ and $S^2 = SS_T/(N-1)$. Then

$$\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = \frac{1}{k}\sum_i(\bar{y}_i-\mu)^2 = \frac{SS_B}{N},$$

and equating with the ρ form, $\frac{SS_T}{nN}[1+(n-1)\rho] = \frac{SS_B}{N}$, gives

$$1+(n-1)\rho = \frac{n\,SS_B}{SS_T} \quad\Longrightarrow\quad \rho = 1-\frac{n}{n-1}\frac{SS_W}{SS_T} \approx 1-\frac{SS_W}{SS_T} = \frac{SS_B}{SS_T}.$$

A large between-sample sum of squares means a large ρ and a large variance.

Example 4.9: Two orderings of $1, \dots, 9$

Here $\mu = 5$, $S^2 = 60/8 = 7.5$, $n = k = 3$, and the SRS variance is $\mathrm{Var}(\bar{y}_{\mathrm{ran}}) = \frac{7.5}{3} \cdot \frac{6}{9} = 1.667$.

- Order 1 2 3 | 4 5 6 | 7 8 9: the samples are $(1,4,7), (2,5,8), (3,6,9)$ with means $4, 5, 6$. Then $\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = \frac{1}{3}(1+0+1) = \frac{2}{3}$ and $\rho = -21/60$. Heterogeneous samples: systematic beats SRS.
- Order 1 4 7 | 2 5 8 | 3 6 9: the samples are $(1,2,3), (4,5,6), (7,8,9)$ with means $2, 5, 8$. Then $\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = \frac{1}{3}(9+0+9) = 6$ and $\rho = 51/60$. Homogeneous samples: systematic is much worse than SRS.

Check: $\frac{7.5}{3} \cdot \frac{8}{9}(1+2\rho) = \frac{20}{9}(1+2\rho)$ equals $\frac{2}{3}$ for $\rho = -0.35$ and 6 for $\rho = 0.85$.

3.5 Types of population

We want systematic samples with heterogeneous units (small ρ). Which populations produce them?

1. **Random order**, e.g. weights of $N = 3000$ freshmen from an alphabetical list: $\rho \approx 0$, $\mathrm{Var}(\bar{y}_{\mathrm{sy}}) \approx \mathrm{Var}(\bar{y}_{\mathrm{ran}})$; treat as SRS.
2. **Ordered population**, e.g. farms by area for corn yield: the sample covers the range, $\mathrm{Var}(\bar{y}_{\mathrm{sy}}) < \mathrm{Var}(\bar{y}_{\mathrm{ran}})$, and is more representative than an SRS.
3. **Periodic variation**, e.g. supermarket sales by weekday: if k matches the period, ρ is large and $\mathrm{Var}(\bar{y}_{\mathrm{sy}}) > \mathrm{Var}(\bar{y}_{\mathrm{ran}})$.

3.6 Estimating the variance

One systematic sample carries no between-sample information, so $\mathrm{Var}(\bar{y}_{\mathrm{sy}})$ has no unbiased estimator; assuming random order ($\rho \approx 0$), use the SRS formula:

$$\bar{y}_{\mathrm{sy}} = \frac{1}{n}\sum_{i=1}^{n} y_i, \qquad \widehat{\mathrm{Var}}(\bar{y}_{\mathrm{sy}}) = \frac{s^2}{n} \cdot \frac{N-n}{N}.$$

This is conservative for an ordered population and anti-conservative under periodicity.

4 Population Total and Proportion

Assuming a randomly ordered population:

$$\hat{\tau} = N\bar{y}_{\mathrm{sy}}, \qquad \widehat{\mathrm{Var}}(\hat{\tau}) = N^2\widehat{\mathrm{Var}}(\bar{y}_{\mathrm{sy}}) = N^2\frac{s^2}{n}\frac{N-n}{N};$$

$$\hat{p}_{\mathrm{sy}} = \bar{y}_{\mathrm{sy}} = \frac{1}{n}\sum y_i \ (y_i \in \{0,1\}), \qquad \widehat{\mathrm{Var}}(\hat{p}_{\mathrm{sy}}) = \frac{\hat{p}_{\mathrm{sy}}\hat{q}_{\mathrm{sy}}}{n-1} \cdot \frac{N-n}{N}.$$

5 Selecting the Sample Size

Under the same assumption the SRS formulas apply: for μ with bound B,

$$n = \frac{N\sigma^2}{(N-1)D + \sigma^2} = \frac{NS^2}{ND + S^2}, \qquad D = \frac{B^2}{4},$$

and for p, $n = \dfrac{Npq}{(N-1)D + pq}$ with $D = B^2/4$.

6 Selected exercises

Motivation. Systematic sampling within strata, the exact gain in an ordered population, and a total under random order.

Exercise 4.1 (Systematic sampling within strata)

N elements in L strata of sizes $N_\ell = n_\ell k$ (common k); independent 1-in-k systematic samples (Method A) in each. With $\bar{y}_\ell$ the stratum sample mean and $\hat{\mu} = \sum_\ell w_\ell \bar{y}_\ell$, $w_\ell = N_\ell/N$, find $\mathrm{E}(\hat{\mu})$ and $\mathrm{Var}(\hat{\mu})$.

Solution. Let $y_{\ell ij}$ be element j of possible sample i in stratum ℓ, $\bar{y}_{\ell i} = \frac{1}{n_\ell}\sum_j y_{\ell ij}$, and μ_ℓ, S_ℓ^2, ρ_ℓ the stratum mean, variance and intraclass correlation.

Step 1. (Unbiasedness.) In stratum ℓ the sample i is chosen with probability $1/k$, so by Case 1 of the unbiasedness subsection, $\mathrm{E}(\bar{y}_\ell) = \frac{1}{k}\sum_i \bar{y}_{\ell i} = \frac{1}{N_\ell}\sum_i\sum_j y_{\ell ij} = \mu_\ell$. Hence $\mathrm{E}(\hat{\mu}) = \sum_\ell w_\ell \mu_\ell = \frac{1}{N}\sum_\ell \tau_\ell = \mu$.

Step 2. (Variance.) The random starts are independent across strata, so the covariances vanish:

$$\mathrm{Var}(\hat{\mu}) = \sum_{\ell=1}^{L} w_\ell^2 \mathrm{Var}(\bar{y}_\ell), \qquad \mathrm{Var}(\bar{y}_\ell) = \frac{1}{k}\sum_{i=1}^{k}(\bar{y}_{\ell i} - \mu_\ell)^2.$$

Step 3. (Intraclass form.) Applying the intraclass-correlation theorem within stratum ℓ,

$$\mathrm{Var}(\hat{\mu}) = \sum_{\ell=1}^{L} w_\ell^2 \frac{S_\ell^2}{n_\ell}\cdot\frac{N_\ell - 1}{N_\ell}\big[1 + (n_\ell - 1)\rho_\ell\big].$$

Step 4. (Interpretation.) $n_\ell = N_\ell/k$: automatic proportional allocation with fraction $1/k$. In random order ($\rho_\ell \approx -1/(N_\ell - 1)$) the bracket times $\frac{N_\ell - 1}{N_\ell}$ is $1 - \frac{1}{k}$, giving proportional stratified sampling. Variances are estimated by the SRS formula per stratum. ■

Exercise 4.2 (Population with a linear trend)

Let $y_t = a + bt$, $t = 1, \ldots, N$, with $b \neq 0$ and $N = nk$, $n \geq 2$. Show that

$$\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = \frac{b^2(k^2-1)}{12}, \qquad \mathrm{Var}(\bar{y}_{\mathrm{ran}}) = \frac{b^2(k-1)(N+1)}{12}, \qquad \frac{\mathrm{Var}(\bar{y}_{\mathrm{sy}})}{\mathrm{Var}(\bar{y}_{\mathrm{ran}})} = \frac{k+1}{N+1} < 1.$$

Solution. [Supplement]

Step 1. (A discrete uniform variance.) If I is uniform on $\{1, \ldots, k\}$, then $\mathrm{E}I = \frac{k+1}{2}$, $\mathrm{E}I^2 = \frac{(k+1)(2k+1)}{6}$ and

$$\mathrm{Var}I = \frac{(k+1)(2k+1)}{6} - \frac{(k+1)^2}{4} = \frac{(k+1)\big[2(2k+1) - 3(k+1)\big]}{12} = \frac{k^2-1}{12}.$$

Step 2. (Systematic mean.) Sample i consists of $t = i, i+k, \ldots, i+(n-1)k$, so $\bar{y}_i = a + b\big(i + \frac{(n-1)k}{2}\big)$. The start I is uniform on $\{1, \ldots, k\}$, so $\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = b^2\mathrm{Var}I = \frac{b^2(k^2-1)}{12}$.

Step 3. (SRS.) Step 1 with k replaced by N gives $\sum_{t=1}^{N}\big(t - \frac{N+1}{2}\big)^2 = N\frac{N^2-1}{12}$, so $S^2 = \frac{b^2}{N-1}\cdot\frac{N(N^2-1)}{12} = \frac{b^2N(N+1)}{12}$. With $1 - f = 1 - \frac{1}{k}$ and $N = nk$,

$$\mathrm{Var}(\bar{y}_{\mathrm{ran}}) = \frac{k-1}{k}\cdot\frac{1}{n}\cdot\frac{b^2N(N+1)}{12} = \frac{b^2(k-1)(N+1)}{12}.$$

Step 4. (Ratio.) Dividing, $\frac{(k^2-1)}{(k-1)(N+1)} = \frac{k+1}{N+1}$, which is below 1 because $k < N$ when $n \geq 2$. For $N = 1000$ and $k = 10$ the systematic sample has about 1% of the SRS variance: it cannot pick only small or only large units, which is the "ordered population" case of the text made exact. ■

Exercise 4.3 (Families who rent)

A 1-in-50 systematic sample of tax records ($N = 15{,}200$ households) estimates the number of renting families: of $n = 304$, $\sum y_i = 88$ rent. Estimate the total with a bound, assuming random order with respect to renting.

Solution.

Step 1. $\hat{p}_{\text{sy}} = 88/304 = 0.2895$ and $\hat{\tau} = N\hat{p}_{\text{sy}} = 15{,}200 \times 0.2895 = 4400$.

Step 2. Under the random-order assumption the SRS formulas apply:

$$\widehat{\text{Var}}(\hat{p}_{\text{sy}}) = \frac{\hat{p}\hat{q}}{n-1} \cdot \frac{N-n}{N} = \frac{0.2895 \times 0.7105}{303} \cdot \frac{14{,}896}{15{,}200} = 0.000665.$$

Step 3. $\widehat{\text{Var}}(\hat{\tau}) = 15{,}200^2 \times 0.000665 = 153{,}700$, $2\sqrt{153{,}700} = 784$: 4400 ± 784 families, valid only without periodicity of period 50 related to renting (e.g. apartment blocks of 50). ■